

Aslesha Mohanty

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Professional Summary

CSE undergraduate with experience building data-intensive applications and analytics platforms. Experience translating ambiguous business questions into SQL-driven insights and production-ready data workflows. Designed ETL pipelines and SQL-based analyses on real-world datasets to generate insights and support product decisions.

Education

Manipal Institute of Technology

July 2023 – July 2027

BTech (Honours) Computer Science and Engineering (Data Science) | CGPA: 8.48/10

Relevant Coursework: DSA, DBMS, OOPs, Math for ML, NLP, Big Data Analytics

Experience

Backend Intern, Karkhana

May 2025 – September 2025

- Built and maintained backend services using Python and Frappe, designing REST APIs and database schemas to support manufacturing workflows and internal tooling.
- Developed a feature detection algorithm in FreeCAD for automated CAD model analysis, improving design validation accuracy and reducing manual inspection overhead.

Projects

User Engagement Optimization & Churn Prediction System | Product Analytics, Data Science, ML, SHAP, Clustering

- Performed funnel analysis and cohort retention analysis to identify engagement drop-offs and early churn patterns.
- Engineered behavioral and lifecycle features to train churn models (Logistic Regression, Random Forest, XGBoost), achieving ROC-AUC of 0.84.
- Applied SHAP explainability to uncover key churn drivers and generate user-level insights for targeted retention strategies.

Seller Performance Intelligence Platform | Python, SQL, FastAPI, PostgreSQL, ETL, Streamlit

- Built a production-style BI platform using the Olist e-commerce dataset, transforming multi-table transactional data into an analytics-ready relational schema.
- Developed advanced SQL analytics using window functions and time-series aggregations to compute KPIs including On-Time Delivery Rate, Order Defect Rate, and Revenue Growth.

ThreatSequence | FastAPI, React, AWS EC2, Nginx, Google OAuth

- Built and deployed an AI-powered cybersecurity threat awareness platform with integrated blog management for a startup.
- Designed and implemented an article publishing and management platform with role-based access (admin/user), enabling clients to securely create, edit, and manage content.
- Deployed production infrastructure on AWS EC2 using Nginx reverse proxy, Gunicorn, and HTTPS via Let's Encrypt.

AI-Enabled Audit Assistant for Vulnerability Classification | Python, XGBoost, Security Analytics

- Built an AI-powered audit assistant integrating Fortify and Debricked outputs to classify vulnerabilities as True/False Positives.
- Correlated static, dynamic, and runtime signals to improve exploitability scoring and streamline validated findings into workflows.

Skills and Technologies

Languages: Python, C++, SQL, C

AI/ML: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, SHAP, XGBoost

Data & Tools: Pandas, NumPy, SQL Analytics, ETL Pipelines, Docker, Linux, Hadoop, Matplotlib, Seaborn, OpenCV, NLTK, REST APIs, Git

Web/Backend: Flask, Streamlit, React, PostgreSQL, AWS (EC2, S3), Nginx, FastAPI

Selected Achievements

Top 10/869 (M# Manipal **Nationwide** Hackathon) | Top 25/250+ (Honeywell x MIT Hackathon) | Special Mention (MIT x ACM x Nokia Hackathon) | Top 6/100 (MIT Bengaluru Hackathon)

Publications

Artificial Intelligence in Credit Scoring: Enhancing Financial Inclusion & Opportunities

International Research Journal of Education and Technology (IRJEdT), Vol. 08, Issue 04, April 2025

DOI: 10.5281/zenodo.15256246

Certifications

AWS Academy Machine Learning Foundations — *In Progress*

Python and Statistics for Financial Analysis — HKUST (Coursera)